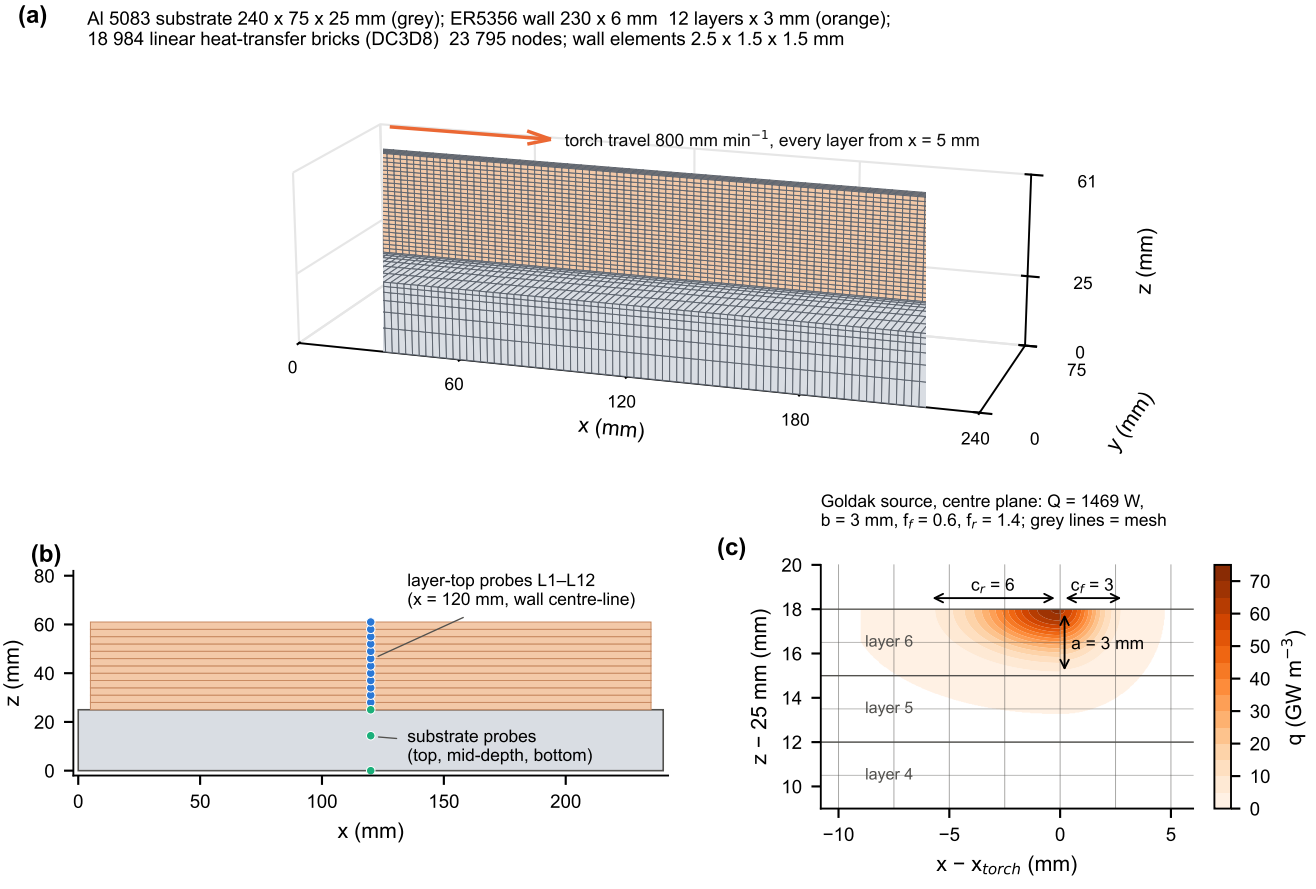

(a)
Al 5083 substrate 240 x 75 x 25 mm (grey); ER5356 wall 230 x 6 mm 12 layers x 3 mm (orange);
18 984 linear heat-transfer bricks (DC3D8) 23 795 nodes; wall elements 2.5 x 1.5 x 1.5 mm
torch travel 800 mm min^{-1}, every layer from x = 5 mm
0
60
120
180
240
x (mm)
61
25
0
z (mm)
0
75
y (mm)
(b)
layer-top probes L1–L12
(x = 120 mm, wall centre-line)
substrate probes
(top, mid-depth, bottom)
z (mm)
x (mm)
Goldak source, centre plane: Q = 1469 W,
b = 3 mm, f_f = 0.6, f_r = 1.4; grey lines = mesh
(c)
c_r = 6
c_f = 3
a = 3 mm
layer 6
layer 5
layer 4
z − 25 mm (mm)
x − x_{torch} (mm)
q (GW m^{-3})